

GENX: Porewater Methane

July 2026 Samples

2026-08-28

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```
##Run Information
```

```
cat("Run Information: Brenden Blakley ") #lets you know what section you're in
```

```
## Run Information: Brenden Blakley
```

```
library(here)
```

```
## here() starts at C:/Users/blkleybd/OneDrive - Smithsonian Institution/Documents/GitHub/GCREW
```

```
#set the run date & user name
```

```
sample_year <- "2026"
```

```
sample_month <- "July"
```

```
user <- "Brenden Blakley"
```

```
#identify the files you want to read in
```

```
#read in as a list to accommodate multiple runs in a month
```

```
GHG_files <- c("Raw Data/20260817_GCREW_Phrag_GENX_formatted.csv", "Raw Data/20260819_GCREW_GENX_2M_forma
```

```
# Define the file path for QAQC log file - NO Need to change just check year
```

```
log_path <- here("Porewater QAQC Logs", "GCREW_CH4_QAQC_Log.csv")
```

```
# Define final path for data be sure to change year and month and run number
```

```
final_path <- "Processed Data/GCREW_GENX_Porewater_CH4_202607.csv"
```

```
#Gas in Water Calculations
```

```
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
```

```
TempC = 25 #Equilibration Temperature (Celsius)
```

```
volume_g = 0.015 #volume of gas (in liters) in syringe when equilibrated
```

```
volume_w = 0.015 #volume of water (in liters) in syringe when equilibrated (should be equal to volume_g
```

```
#record any notes about the run or anything other info here:
```

```
run_notes <- "Ch1-80 and Ch11-80 Not Collected"
```

```
#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"
```

```
cat(run_notes)
```

```
## Ch1-80 and Ch11-80 Not Collected
```

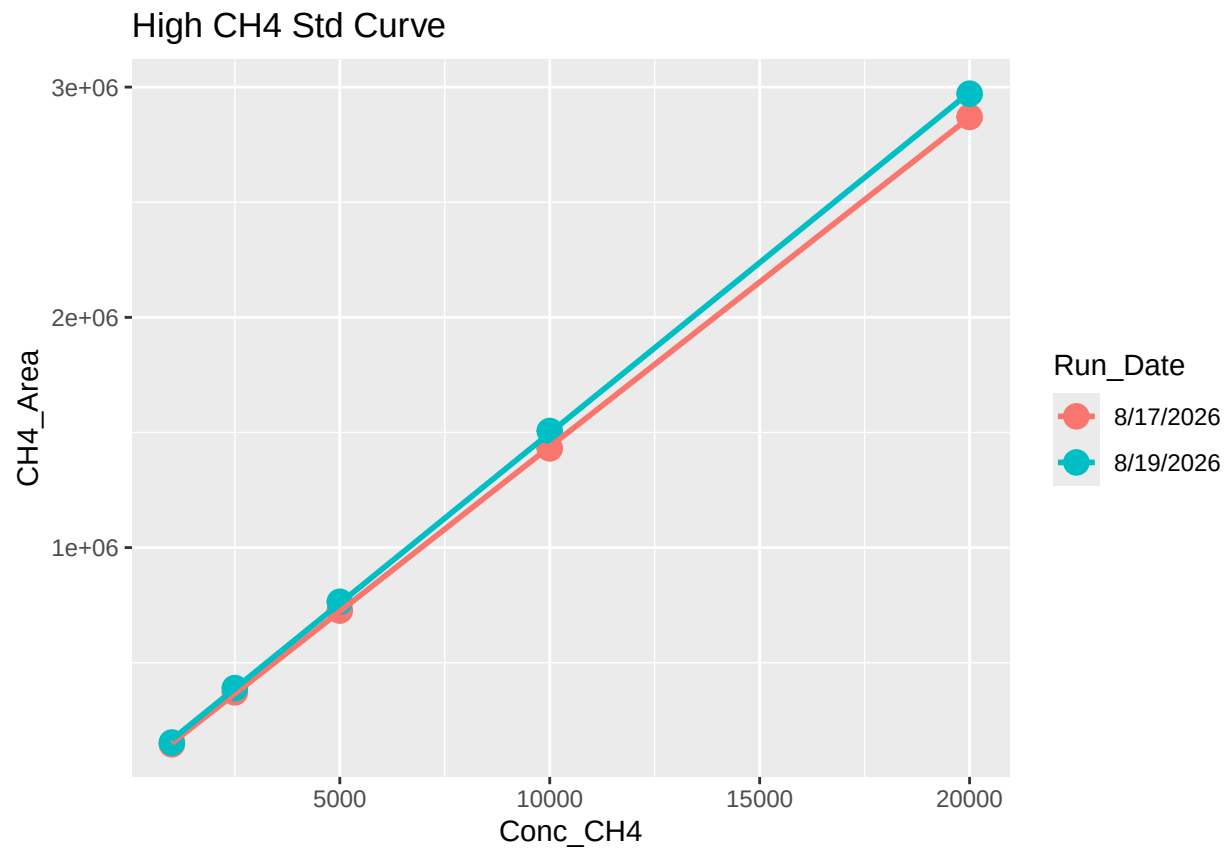
```
##Set Up
```

```
##Read in metadata and create similar sample IDs for matching to samples
```

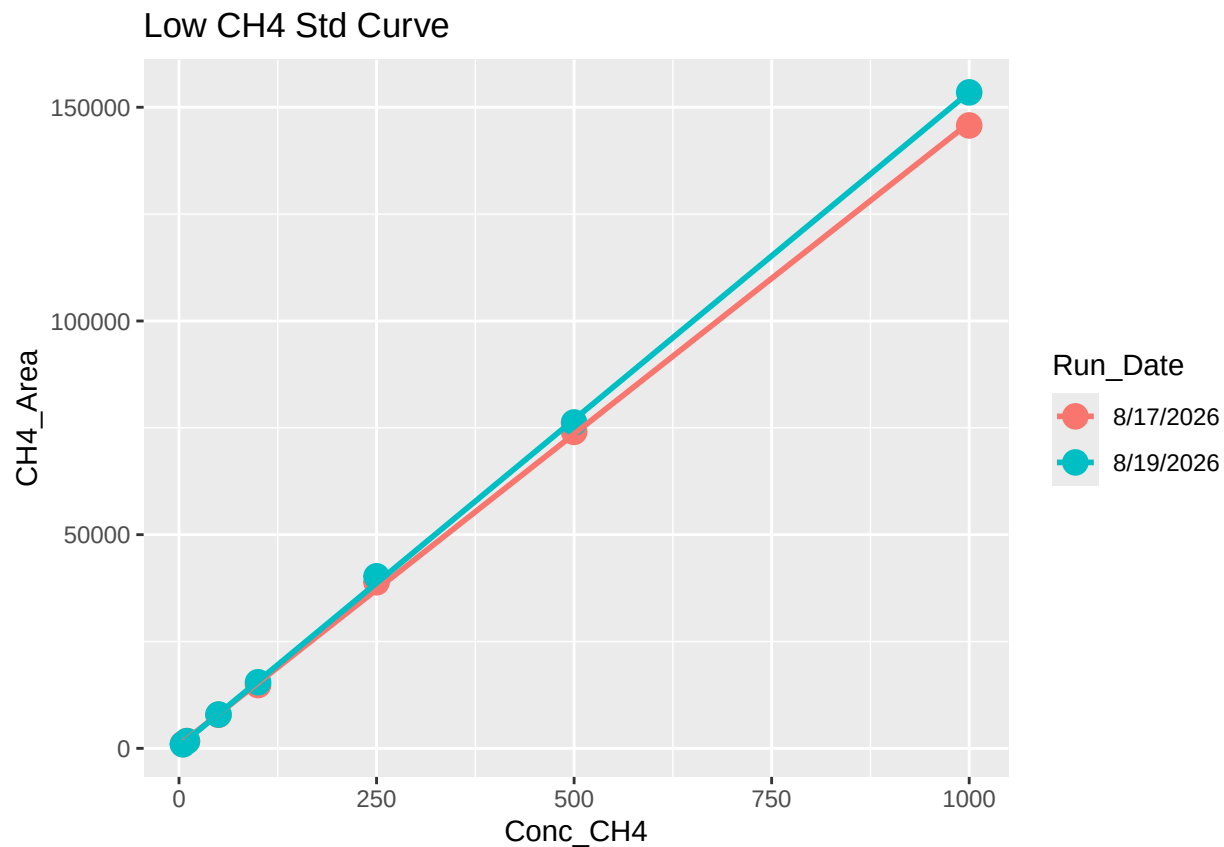
1 Read in data files

2 Assess standard curves

```
## 'geom_smooth()' using formula = 'y ~ x'
```

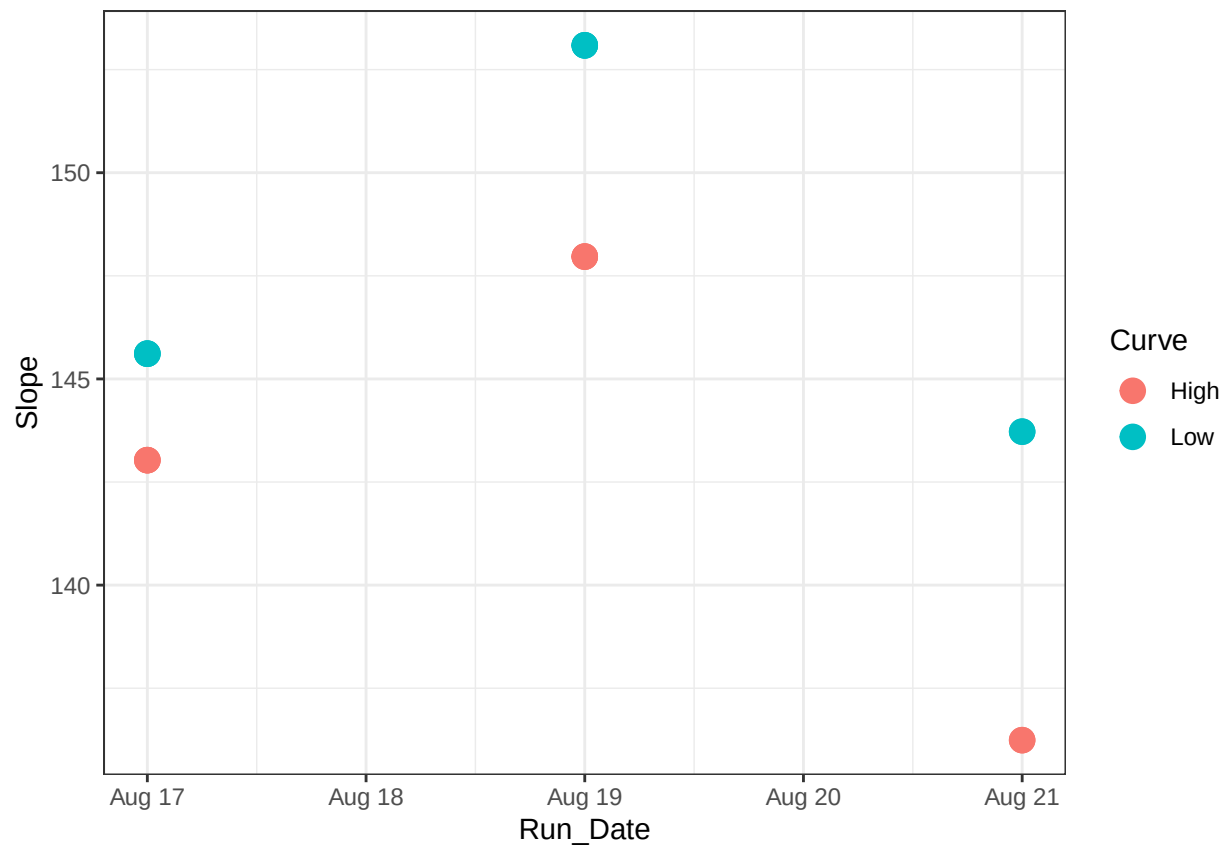


```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## Joining with 'by = join_by(Run_Date)'
```

##	Curve	R2	Slope	Intercept	Run_Date
## 1	High	0.9994458	136.2406	25980.5484	8/21/2026
## 2	Low	0.9995311	143.7214	801.9223	8/21/2026
## 3	High	0.9999168	147.9666	17873.3851	8/19/2026
## 4	Low	0.9997966	153.0888	407.2899	8/19/2026
## 5	High	0.9999646	143.0303	7587.2742	8/17/2026
## 6	Low	0.9996977	145.6125	745.1636	8/17/2026



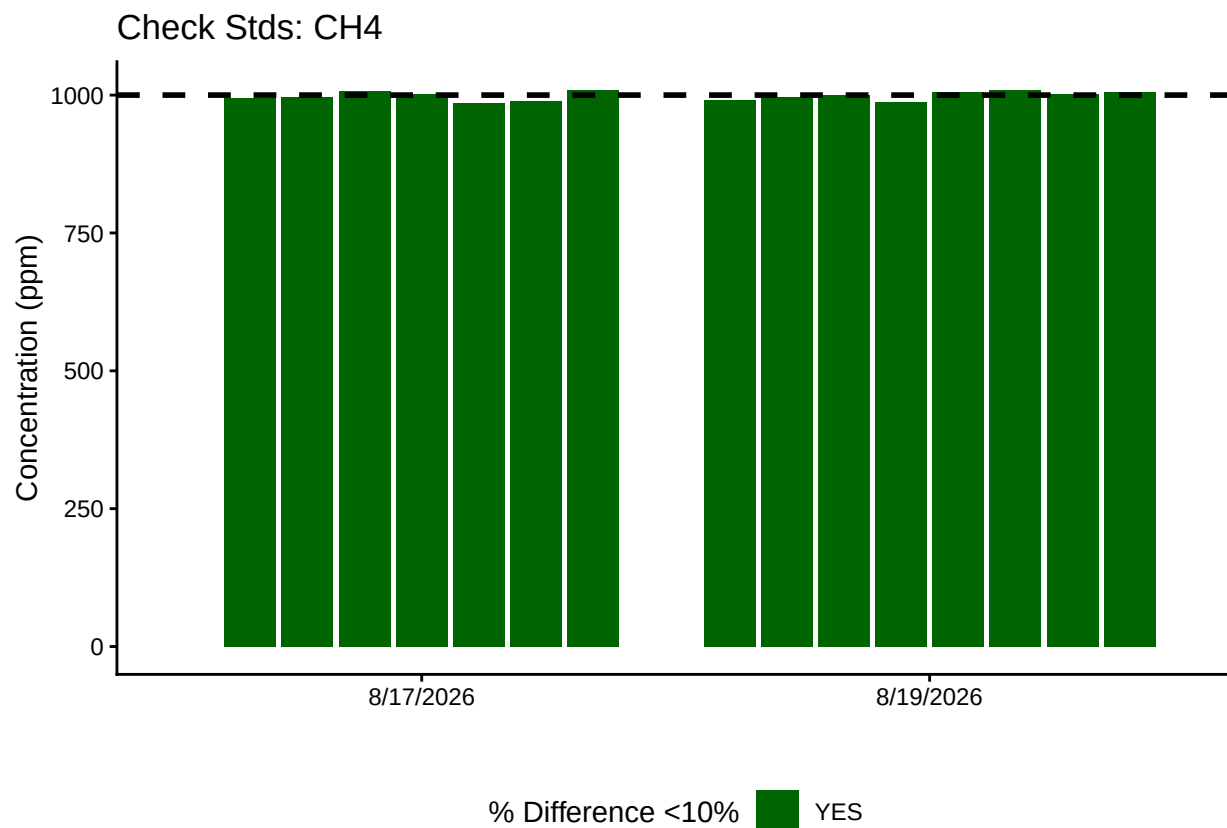
```
## [1] "High_R2 is Good-8/17/2026" "High_R2 is Good-8/19/2026"
```

```
## [1] "Low_R2 is Good-8/17/2026" "Low_R2 is Good-8/19/2026"
```

2.1 Now calculate the CH4 in ppm

3 Aanlayze the Check Standards

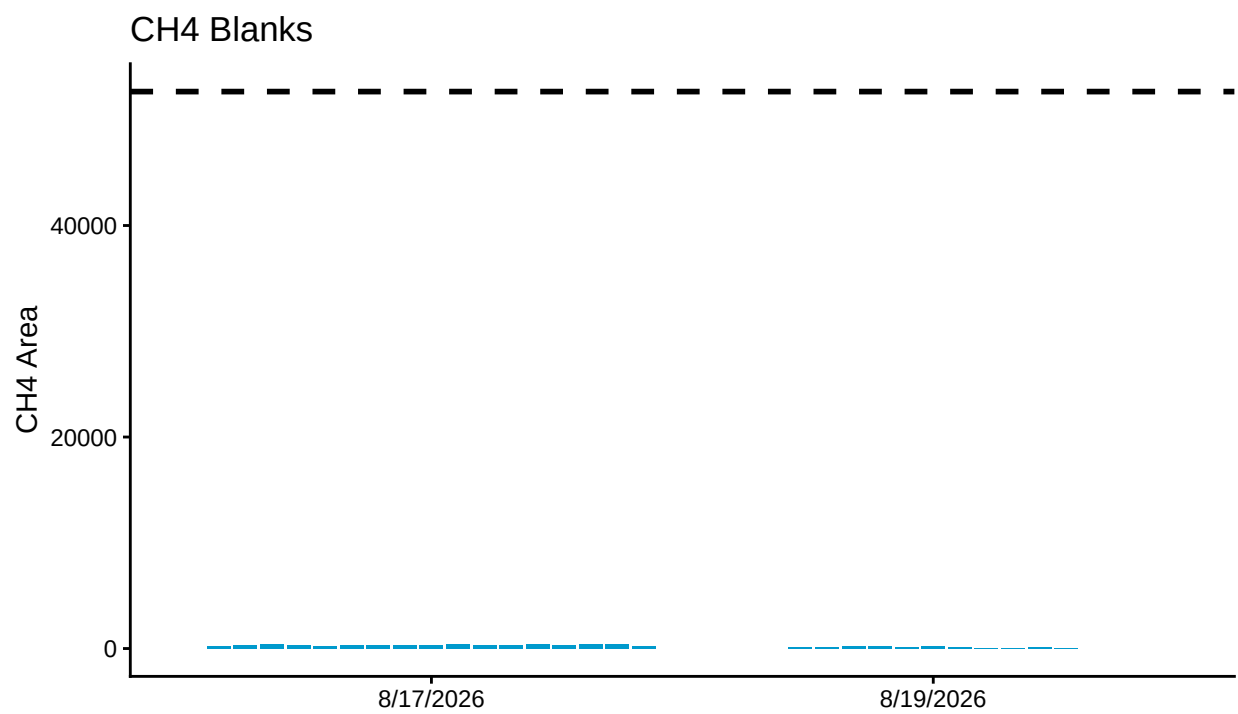
```
## >60% of CH4 Check Standards are within range of expected concentration - PROCEED
```



4 Analyze the Blanks

Assess Blanks

>60% of CH4 Blanks are below the lower 25% quartile of samples - PROCEED

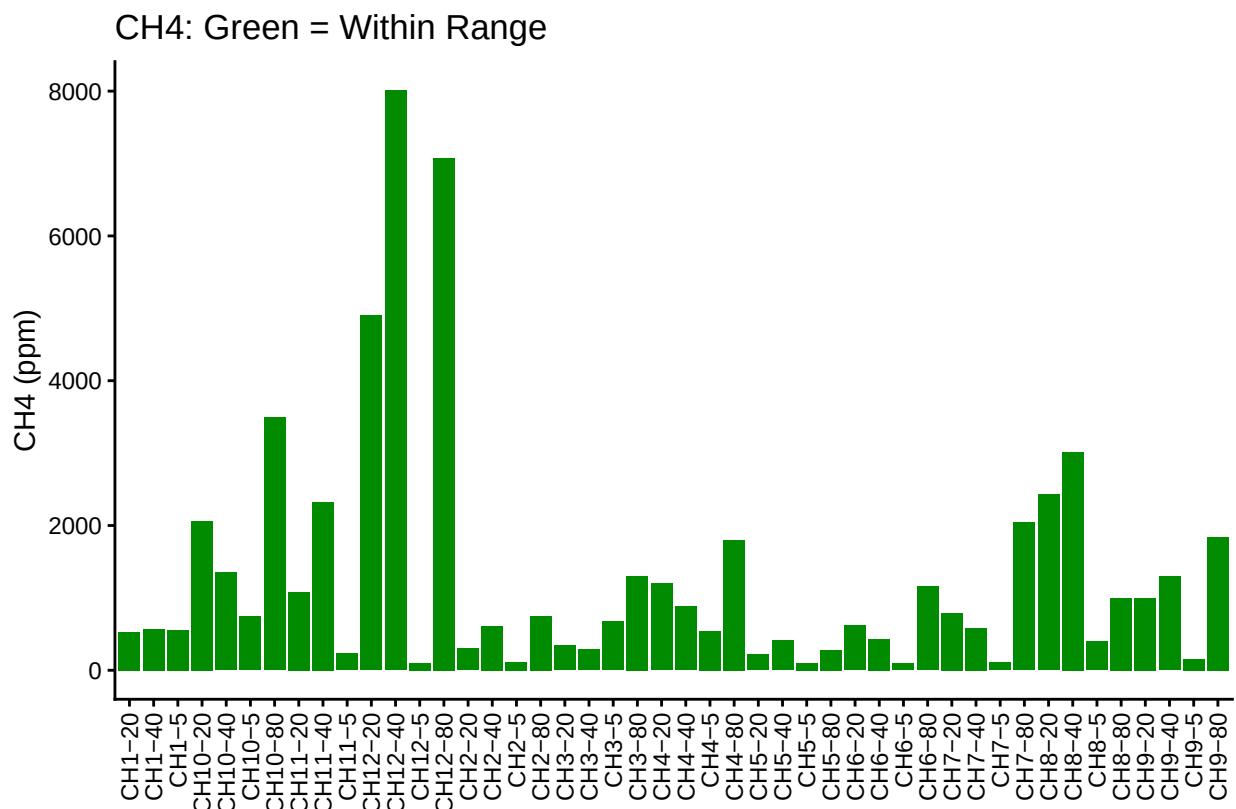
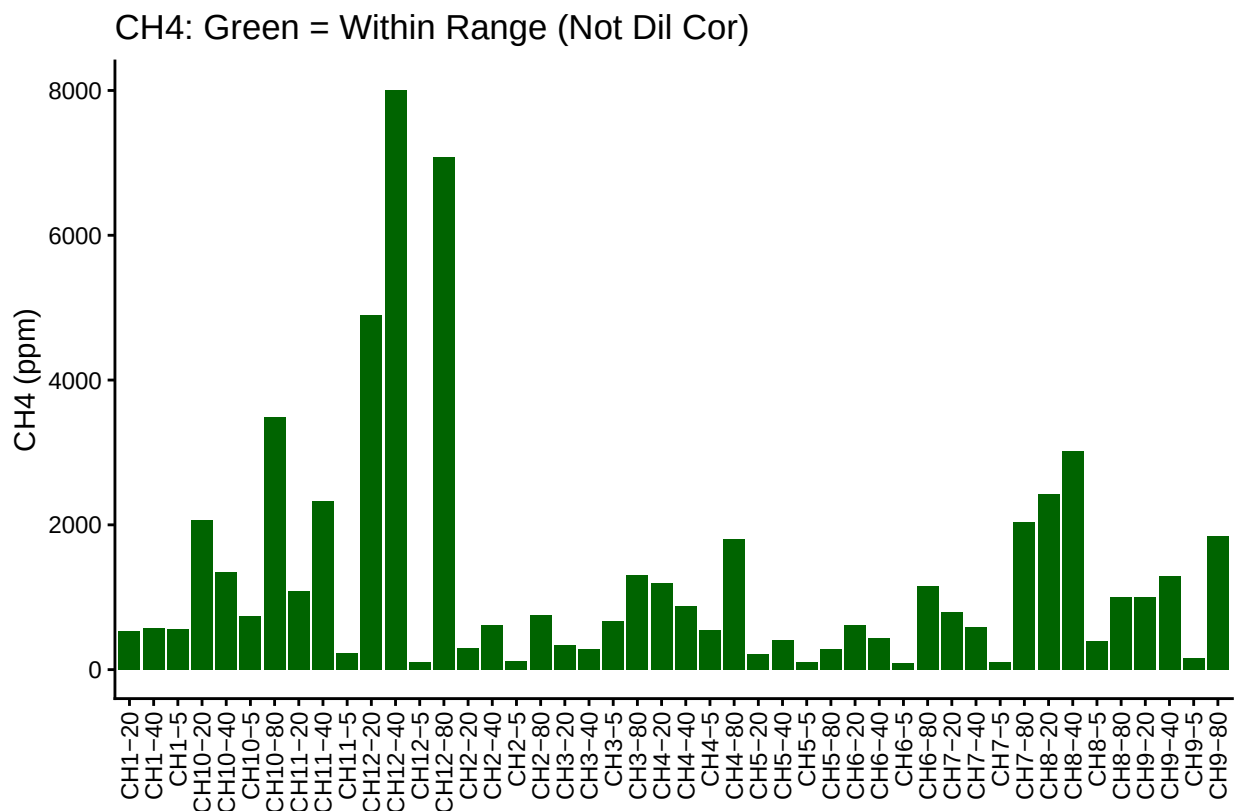


Blank Conc <25% Quartile Samples ■ YES

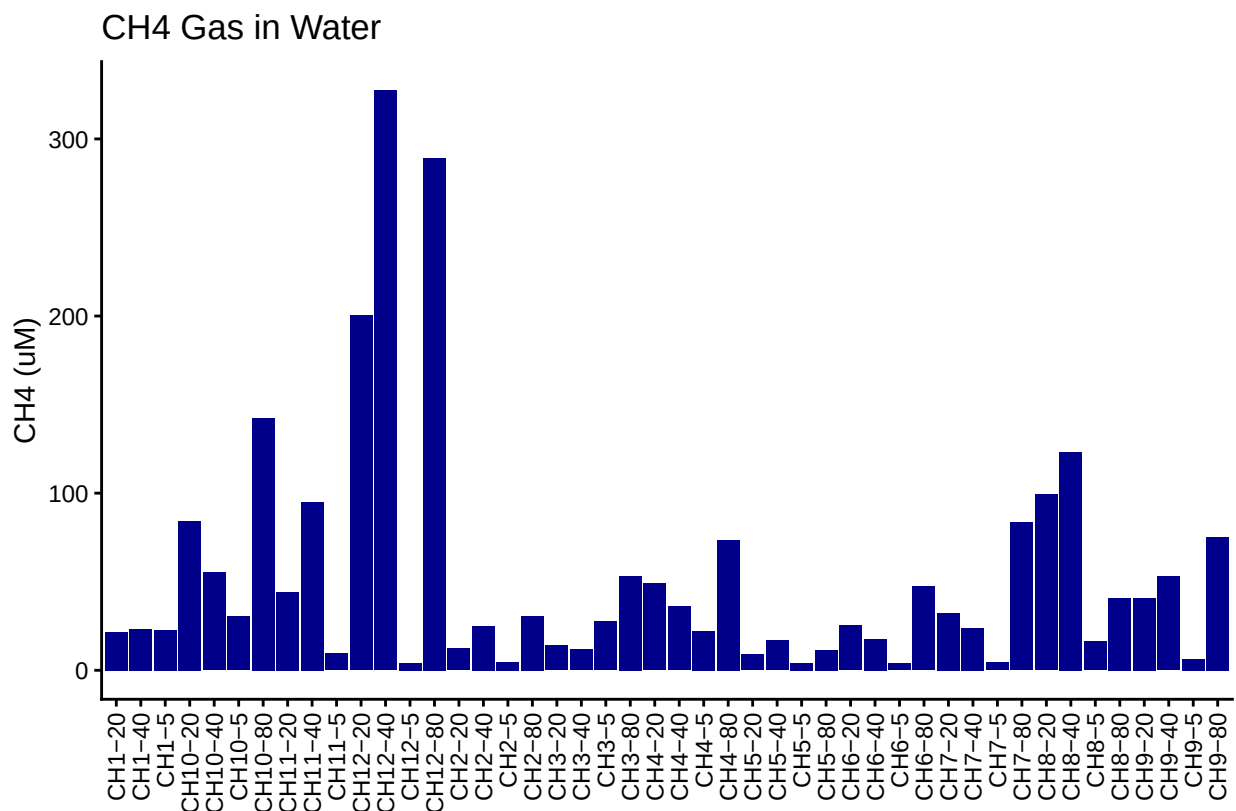
Table 1: Mean Area of Blanks

Test	Blank_Mean_Area
CH4	262.7857

5 Dilution correct samples and Check Range



6 Calculate gas in water



7 Merge samples with Metadata & check all samples are present

```
## Merge Samples with Metadata
```

```
## Some sample IDs are missing from data.
```

```
## [1] "Ch1-80" "Ch11-80"
```

7.1 Organize Data

```
##      Run_Date Curve Sample_ID   Type CH4_Area Conc_CH4      CH4_RTime_Flag
## 1 8/17/2026   Low    Ch1-5 Sample    81695      NA Correct - CH4 RTime
## 2 8/17/2026   Low    Ch1-20 Sample    77253      NA Correct - CH4 RTime
## 3 8/17/2026   Low    Ch1-40 Sample    84141      NA Correct - CH4 RTime
## 4 8/17/2026   Low    Ch2-5 Sample    17049      NA Correct - CH4 RTime
## 5 8/17/2026   Low    Ch2-20 Sample    44315      NA Correct - CH4 RTime
## 6 8/17/2026   Low    Ch2-40 Sample    89828      NA Correct - CH4 RTime
##      CO2_Area Conc_CO2      CO2_RTime_Flag N2O_Area Conc_N2O      N2O_RTime_Flag
## 1   602230      NA Correct - CO2 RTime      211      NA Correct - N2O RTime
## 2   582801      NA Correct - CO2 RTime      244      NA Correct - N2O RTime
```

```

## 3 522770 NA Correct - CO2 RTime 248 NA Correct - N2O RTime
## 4 483189 NA Correct - CO2 RTime 224 NA Correct - N2O RTime
## 5 493125 NA Correct - CO2 RTime 210 NA Correct - N2O RTime
## 6 519609 NA Correct - CO2 RTime 228 NA Correct - N2O RTime
## FID_Peak_Flag ECD_Peak_Flag Sample_Volume R2 Slope Intercept Min
## 1 Correct Correct 12 0.9996977 145.6125 745.1636 5
## 2 Correct Correct 12 0.9996977 145.6125 745.1636 5
## 3 Correct Correct 12 0.9996977 145.6125 745.1636 5
## 4 Correct Correct 12 0.9996977 145.6125 745.1636 5
## 5 Correct Correct 12 0.9996977 145.6125 745.1636 5
## 6 Correct Correct 12 0.9996977 145.6125 745.1636 5
## Max CH4_Conc_ppm CH4_Flag CH4_Conc_ppm_dilcorr CH4_Conc_uM
## 1 20000 555.9266 Within Range 555.9266 22.72224
## 2 20000 525.4210 Within Range 525.4210 21.47539
## 3 20000 572.7246 Within Range 572.7246 23.40882
## 4 20000 111.9673 Within Range 111.9673 4.57641
## 5 20000 299.2178 Within Range 299.2178 12.22985
## 6 20000 611.7803 Within Range 611.7803 25.00513

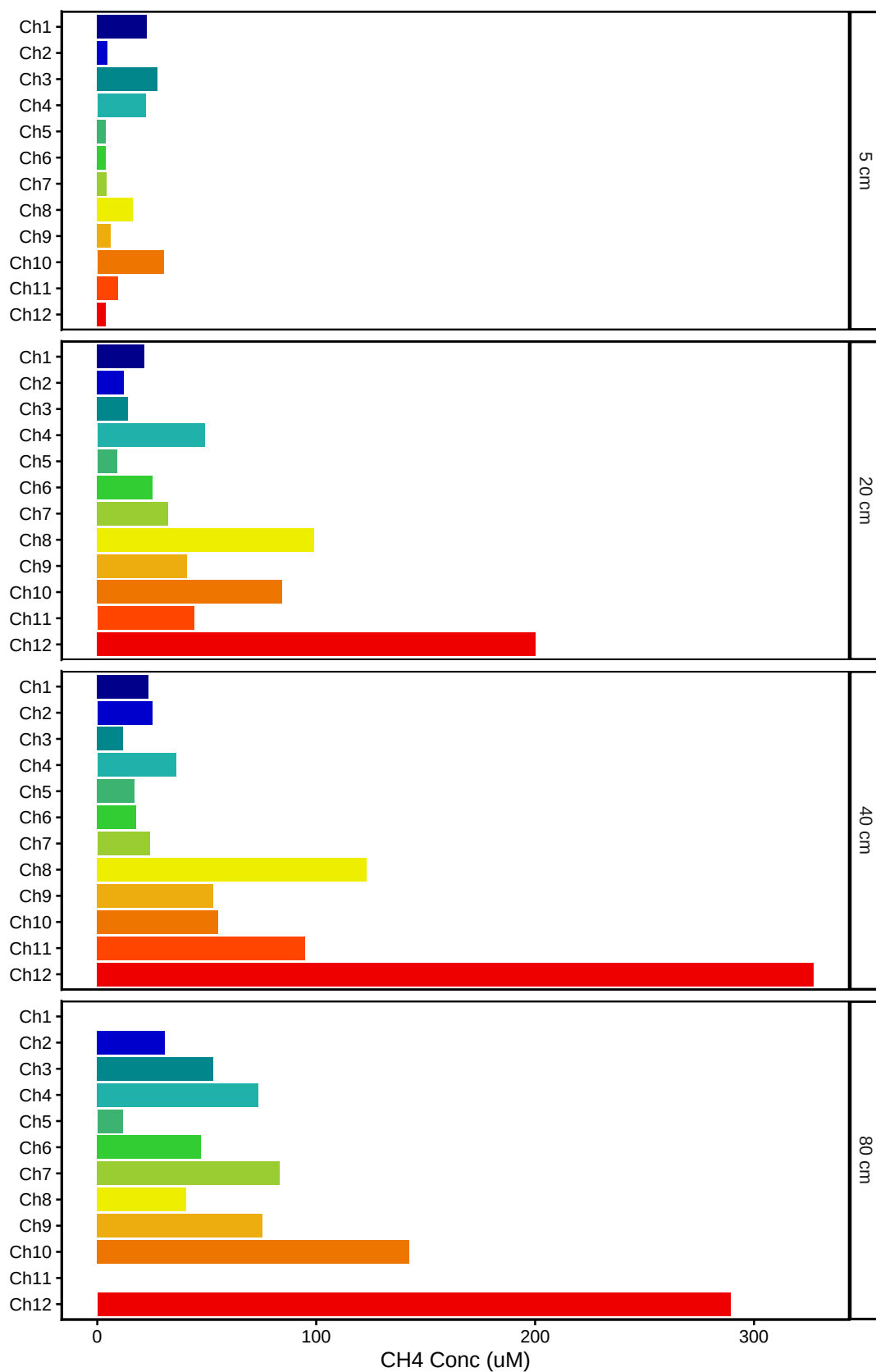
## Site Project Year Month Community Nickname Plot Depth_cm
## 1 GCRew GENX 2026 July SCAM, SPPA, DISP GENX Ch1 5
## 2 GCRew GENX 2026 July SCAM, SPPA, DISP GENX Ch1 20
## 3 GCRew GENX 2026 July SCAM, SPPA, DISP GENX Ch1 40
## 4 GCRew GENX 2026 July SCAM, SPPA, DISP GENX Ch2 5
## 5 GCRew GENX 2026 July SCAM, SPPA, DISP GENX Ch2 20
## 6 GCRew GENX 2026 July SCAM, SPPA, DISP GENX Ch2 40
## CO2_Treatment Temp_Treatment CH4_Conc_ppm CH4_Conc_uM CH4_Conc_Flag
## 1 Ambient 0 555.9266 22.72224 Within Range
## 2 Ambient 0 525.4210 21.47539 Within Range
## 3 Ambient 0 572.7246 23.40882 Within Range
## 4 Ambient 0 111.9673 4.57641 Within Range
## 5 Ambient 0 299.2178 12.22985 Within Range
## 6 Ambient 0 611.7803 25.00513 Within Range
## CH4_QAQC_Flags Analysis_rundate Run_notes
## 1 8/17/2026 Ch1-80 and Ch11-80 Not Collected
## 2 8/17/2026 Ch1-80 and Ch11-80 Not Collected
## 3 8/17/2026 Ch1-80 and Ch11-80 Not Collected
## 4 8/17/2026 Ch1-80 and Ch11-80 Not Collected
## 5 8/17/2026 Ch1-80 and Ch11-80 Not Collected
## 6 8/17/2026 Ch1-80 and Ch11-80 Not Collected

```

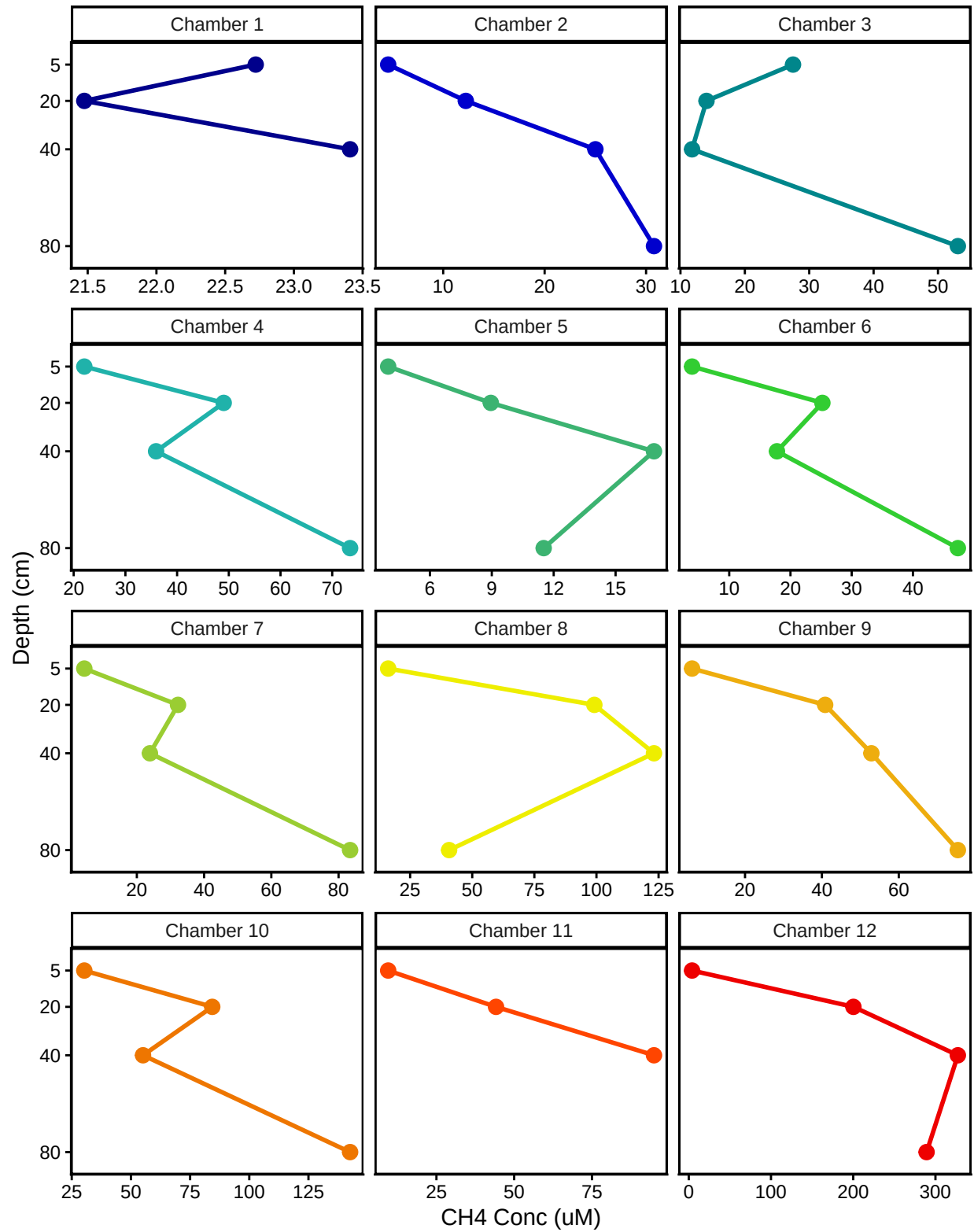
7.2 Visualize Data CH4

```
## Visualize Data
```

GENX: Porewater CH4



GENX: Porewater CH4 by Chamber



##Export Data

#end